

Gym Members Data Analysis using Excel, SQL & Power BI

Project Overview

The Gym Members Data Analysis project focuses on analyzing fitness and workout data using Microsoft Excel, SQL, and Power BI. The objective was to clean and transform raw data, perform exploratory and business-focused analysis, and create an interactive dashboard that provides actionable insights into member fitness performance, workout habits, and calorie expenditure.

The project demonstrates the complete data analysis workflow, from data preparation to visualization, helping gym management understand member behavior and make data-driven decisions.

Project Objectives

- Clean and prepare the dataset for analysis.
- Explore workout patterns and member demographics.
- Identify factors affecting calorie burn.
- Compare workout performance across different groups.
- Build an interactive dashboard for easy decision-making.
- Generate business insights using SQL queries and Power BI visualizations.

Tools & Technologies Used

- Microsoft Excel
 - Data Cleaning
 - Data Validation
 - Removing Duplicates
 - Handling Missing Values
 - Formatting
 - Exploratory Data Analysis
- MySQL
 - Data Exploration
 - Aggregation
 - CASE Statements
 - GROUP BY
 - ORDER BY
 - Business Insight Queries
- Power BI
 - Data Modeling
 - DAX Measures

- KPI Cards
- Interactive Dashboard
- Charts and Visualizations
- Slicers and Filters

Excel Data Cleaning

The dataset was cleaned and prepared before analysis by:

- Removing duplicate records.
- Checking for missing or null values.
- Correcting data types.
- Standardizing column formats.
- Verifying numerical values.
- Ensuring data consistency for analysis.

SQL Analysis

SQL was used to answer business-related questions and generate meaningful insights.

Key Business Questions

- Which workout type burns the most calories?

```
select Workout_Type, avg(Calories_Burned) as Avg_calories_burned
from gym_membersdata
group by Workout_Type
order by Avg_calories_burned desc;
```

Workout_Type	Avg_calories_burned
HIIT	925.8054
Strength	910.6977
Yoga	903.1883
Cardio	884.5137

- Which gender burns more calories on average?

```
select Gender, avg(Calories_Burned) as Avg_calories_burned
from gym_membersdata
group by Gender
order by Avg_calories_burned desc;
```

Gender	Avg_calories_burned
Male	944.4560
Female	862.2489

- Does longer workout duration burn more calories?

```

> select case
  when Session_Duration(hours)<1 then "Less than 1 hour"
  when Session_Duration(hours) between 1 and 1.5 then "1-1.5 hours"
  else "more than 1.5 hours"
end as Duration_group, avg(Calories_Burned) as Avg_calories_burned
from gym_membersdata
group by Duration_group
order by Avg_calories_burned desc;

```

Duration_group	Avg_calories_burned
more than 1.5 hours	1265.3403
1-1.5 hours	903.8809
Less than 1 hour	540.7688

- Which experience level burns the most calories?

```

select Experience_Level, avg(Calories_Burned) as Avg_Calories
from gym_membersdata
group by Experience_Level
order by Avg_Calories desc;

```

Experience_Level	Avg_Calories
3	1265.3403
2	901.9187
1	726.3750

- Which workout type has the highest average heart rate?

```

select Workout_Type, avg(Avg_BPM) as Avg_bpm
from gym_membersdata
group by Workout_Type
order by Avg_bpm desc;

```

Workout_Type	Avg_bpm
Strength	144.3140
Cardio	143.8902
HIIT	143.5249
Yoga	143.2678

- Which age group burns the most calories?

```

select case
  when Age <25 then "18-24"
  when Age between 25 and 34 then "25-34"
  when Age between 35 and 44 then "35-44"
  else "45+"
end as Age_group, avg(Calories_Burned) as AvgCalories
from gym_membersdata
group by Age_group
order by AvgCalories desc;

```

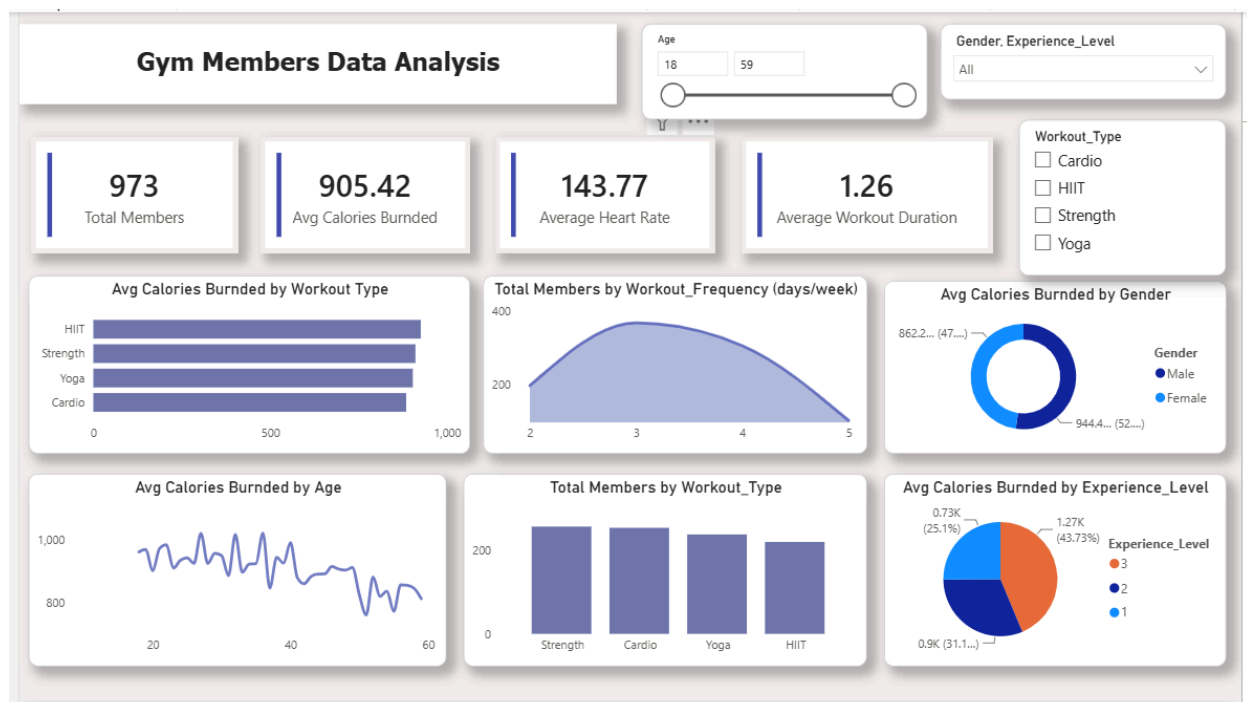
	Age_group	AvgCalories
▶	18-24	949.3394
	25-34	942.9136
	35-44	915.7281
	45+	855.8556

Power BI Dashboard

The Power BI dashboard was designed to present insights in a visually interactive format.

Dashboard Features

- KPI Cards
 - Total Members
 - Average Calories Burned
 - Average Workout Duration
 - Average Heart Rate (BPM)
- Interactive Filters
 - Gender
 - Workout Type
 - Experience Level
 - Age Group
- Charts
 - Calories Burned by Workout Type
 - Calories Burned by Age Group
 - Average Heart Rate by Workout Type
 - Workout Duration Analysis
 - Gender Comparison
 - Experience Level Analysis



Key Insights

- Workout type has a significant impact on calorie expenditure and workout intensity.
- Members participating in longer workout sessions generally burn more calories.
- Experienced members achieve higher average calorie burn due to increased workout intensity.
- Different workout types produce varying average heart rates, indicating different cardiovascular demands.
- Calorie expenditure differs across age groups, reflecting variations in fitness levels and exercise habits.
- Differences in average calories burned between male and female members provide valuable insights into workout patterns.

Business Value

The analysis can help gym owners and fitness professionals:

- Recommend the most effective workout programs.
- Design personalized fitness plans.
- Improve member engagement and retention.
- Optimize training schedules.
- Monitor workout intensity using heart rate analysis.
- Support data-driven decision-making.

Skills Demonstrated

Excel

- Data Cleaning
- Data Validation
- Data Preparation
- Data Formatting

SQL

- Data Querying
- Aggregate Functions
- GROUP BY
- CASE Statements
- Business Analysis
- Data Exploration

Power BI

- Data Modeling
- DAX Measures

- Dashboard Design
- KPI Development
- Interactive Reporting
- Data Visualization

Conclusion

This project demonstrates an end-to-end data analysis workflow using Excel, SQL, and Power BI. Starting with data cleaning in Excel, followed by business analysis in SQL, and concluding with an interactive Power BI dashboard, the project showcases the complete analytical process. The insights generated help understand workout performance, member behavior, and fitness trends while highlighting practical skills in data preparation, querying, visualization, and business intelligence. This project reflects the core competencies expected of a Data Analyst and serves as a strong addition to a professional portfolio.